



Ahmet Fatih Yegen

Software Engineer | Embedded Systems, Robotics & Autonomous Systems

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Summary

Multidisciplinary Software Engineer with a BS in Electrical and Electronics Engineering and intensive C/C++ training from 42 Istanbul. Specialized in embedded systems, industrial robotics integration, and complex Linux environments. Expert in reverse-engineering legacy hardware, structuring Robot Operating System (ROS) frameworks, and designing custom microcontroller platforms (ESP32) utilizing high-precision ADCs and IoT protocols.

Experience

Alpler Ziraat Aletleri A.Ş., Software Development Specialist Engineer – Aydın, Türkiye Jan 2026 – present

- Revitalized and reprogrammed an idle 3-axis KUKA KR210 industrial robot, diagnosing Interbus and servo drive communications to restore operational capacity.
- Designed a comprehensive server-side integration utilizing Robot Operating System (ROS) nodes, Gazebo simulations, and camera arrays for autonomous collision-evasion tracking.
- Structured external server-side Python/C++ frameworks to safely handle machine vision processing and real-time trajectory data over network sockets.
- Maintained multi-boot development setups (Ubuntu/Kali) and deployed custom NMCLI network split-routing policies to manage local camera subnets simultaneously with global traffic.

Alpler Ziraat Aletleri A.Ş., Automation and Maintenance Development Engineer – Aydın, Türkiye Jan 2023 – Aug 2023

- Programmed industrial PLC and HMI automation systems utilizing ISPsoft, WPLsoft, and DOPSoft platforms for heavy machinery.
- Drafted and evaluated complex electrical schematics, Piping and Instrumentation Diagrams (P&ID), and electronic circuit blueprints.
- Executed advanced diagnostics on automated production lines, reducing operational downtime by 15% through preventative logic adjustments.

Port Kargo Ve Lojistik A.Ş., Software Development Specialist – Istanbul, Türkiye Sept 2023 – Jan 2025

- Integrated high-performance 1D/2D Datamatrix barcode scanning modules into mobile applications, boosting warehouse hardware processing speeds by 25%.
- Automated legacy project migrations to Git repositories using custom PowerShell scripts and accelerated EPL barcode processing pipelines via WSL Bash.
- Architected backend logistics infrastructure utilizing .NET Core, N-Tier, and Onion Architecture to manage high-throughput database traffic.

Projects

Custom Microcontroller Battery Telemetry Platform Mar 2026 – present

- Engineered a modular DC internal resistance (IR) testing device centered around ESP8266/ESP32 architectures with a 4-wire Kelvin connection differential topology.
- Designed automated voltage attenuation networks to scale inputs up to 100V DC, isolating control circuitry via optocouplers and utilizing an external ADS1115 16-bit ADC.
- Authored type-safe C++ template dispatchers, custom LittleFS file schemas, and global struct variables to link diagnostics displays securely with Wi-Fi logs.

Education

Cert. 42 Istanbul, Cert. in Intensive Software Engineering Curriculum – Istanbul, Türkiye Nov 2023 – Nov 2025

- Mastered rigorous peer-to-peer software development projects specializing in C/C++ programming, UNIX memory management, low-level architecture, and BASH scripting.

BS Pamukkale University, BS in Electric and Electronics Engineering – Denizli, Türkiye Sept 2014 – Dec 2019

- Developed a semi-supervised machine learning model utilizing Advanced Optimization Techniques for predictive data analysis.

Exch. Warsaw University Of Technology, Exch. in Electrical Engineering & Electronics and Information Technology – Warsaw, Poland 2016-09 - 2017-06

Skills

Systems & Low-Level: C, C++, Python, BASH, PowerShell, UNIX/Linux (Ubuntu, Kali, Distrobox)

Robotics & Automation: OpenCV, deepface, yolov8, KUKA Robotics, PLC Programming (ISPsoft), P&ID

Embedded & Hardware: ESP32/NodeMCU, ADS1115, I2C, MQTT, Modbus, Custom PCB Layouts, Circuit Diagnostics

Backend Architecture: .NET Core, SQL, RESTful APIs, Docker, Git, WSL, Microservices